Sequences & Series

ANSWER KEY

Arithmetic sequences: finding terms

1 Find the 12th term of the arithmetic sequence with $a = 5$, $d = 3$.

$a_{12} = 5 + 11(3) = 5 + 33 = 38.$

2 Find the 20th term of the arithmetic sequence with $a = -4$, $d = 2$.

$a_{20} = -4 + 19(2) = -4 + 38 = 34.$

3 Find the 15th term of the arithmetic sequence with $a = 100$, $d = -6$.

$a_{15} = 100 + 14(-6) = 100 - 84 = 16.$

4 Write the first four terms of the arithmetic sequence with $a = 7$, $d = -2$.

$7,\ 5,\ 3,\ 1.$

Arithmetic sequences: finding a and d

5 The 4th term of an arithmetic sequence is 17 and the 9th term is 42. Find the first term a and common difference d.

$a + 3d = 17$ and $a + 8d = 42$. Subtracting: $5d = 25 \Rightarrow d = 5$. Then $a = 17 - 15 = 2$.

6 An arithmetic sequence has $a_1 = 10$ and $a_6 = 30$. Find the common difference d.

$a_6 = a_1 + 5d = 10 + 5d = 30 \Rightarrow 5d = 20 \Rightarrow d = 4.$

7 An arithmetic sequence has $a_3 = 11$ and $a_7 = 27$. Find the common difference d.

$a_7 - a_3 = 4d = 16 \Rightarrow d = 4.$

Arithmetic series: the sum formula

8 Find the sum of the first 20 terms of the arithmetic sequence with $a = 3$, $d = 4$, using $S_n = \frac{n}{2}(2a + (n-1)d)$.

$S_{20} = \frac{20}{2}(6 + 19(4)) = 10(6 + 76) = 10(82) = 820.$

9 Find the sum of the first 15 terms of the arithmetic sequence with $a = 50$, $d = -3$.

$S_{15} = \frac{15}{2}(100 + 14(-3)) = 7.5(100 - 42) = 7.5(58) = 435.$

10 Find the sum of the arithmetic series $2 + 5 + 8 + \cdots + 50$.

Here $a = 2$, $d = 3$; $50 = 2 + (n-1)3 \Rightarrow n = 17$. $S_{17} = \frac{17}{2}(2 + 50) = 17(26) = 442.$

11 Find the sum of the first 10 positive multiples of 6.

These form an arithmetic sequence with $a = 6$, $d = 6$, $n = 10$.

$S_{10} = \frac{10}{2}(12 + 9(6)) = 5(12 + 54) = 5(66) = 330.$

Geometric sequences: finding terms

12 Find the 6th term of the geometric sequence with $a = 3$, $r = 2$.

$a_6 = 3(2)^5 = 3(32) = 96.$

13 Find the 5th term of the geometric sequence with $a = 100$, $r = 0.5$.

$a_5 = 100(0.5)^4 = 100(0.0625) = 6.25.$

14 Find the 4th term of the geometric sequence with $a = 2$, $r = -3$.

$a_4 = 2(-3)^3 = 2(-27) = -54.$

15 Write the first four terms of the geometric sequence with $a = 5$, $r = 3$.

$5,\ 15,\ 45,\ 135.$

Geometric sequences: finding a and r

16 The 2nd term of a geometric sequence is 6 and the 5th term is 162. Find r and a.

$ar = 6$ and $ar^4 = 162$. Dividing: $r^3 = 27 \Rightarrow r = 3$. Then $a = 6/3 = 2$.

17 A geometric sequence has $a_1 = 4$ and $a_4 = 108$. Find r.

$a_4 = a_1 r^3 = 4r^3 = 108 \Rightarrow r^3 = 27 \Rightarrow r = 3.$

18 A geometric sequence has 3rd term 12 and 6th term 96. Find a and r.

$ar^2 = 12$ and $ar^5 = 96$. Dividing: $r^3 = 8 \Rightarrow r = 2$. Then $a = 12/4 = 3$.

Geometric series: the finite sum

19 Find the sum of the first 6 terms of the geometric series with $a = 3$, $r = 2$, using $S_n = \frac{a(r^n - 1)}{r - 1}$.

$S_6 = \frac{3(2^6 - 1)}{2 - 1} = 3(63) = 189.$

20 Find the sum of the first 5 terms of the geometric series with $a = 100$, $r = 0.5$.

$S_5 = \frac{100(1 - 0.5^5)}{1 - 0.5} = \frac{100(0.96875)}{0.5} = 193.75.$

21 Find the sum of the first 4 terms of the geometric series with $a = 2$, $r = -3$.

$S_4 = \frac{2(1 - (-3)^4)}{1 - (-3)} = \frac{2(1 - 81)}{4} = \frac{-160}{4} = -40.$

Infinite geometric series

22 Find the sum of the infinite geometric series with $a = 4$, $r = \frac{1}{2}$.
$S = \frac{a}{1-r} = \frac{4}{1-\frac{1}{2}} = 8$.

23 Find the sum of the infinite geometric series with $a = 9$, $r = \frac{1}{3}$.
$S = \frac{9}{1-\frac{1}{3}} = \frac{9}{2/3} = 13.5$.

24 Find the sum of the infinite geometric series $8 - 4 + 2 - 1 + \cdots$.
Here $a = 8$, $r = -\frac{1}{2}$. $S = \frac{8}{1-(-\frac{1}{2})} = \frac{8}{1.5} = \frac{16}{3}$.

25 Express the repeating decimal $0.333\ldots$ as a fraction using an infinite geometric series with $a = \frac{3}{10}$, $r = \frac{1}{10}$.
$S = \frac{3/10}{1-1/10} = \frac{3/10}{9/10} = \frac{3}{9} = \frac{1}{3}$.

Sigma notation

26 Evaluate $\sum_{k=1}^{5} (2k+1)$.
Terms: $3, 5, 7, 9, 11$. Sum $= 35$.

27 Evaluate $\sum_{k=1}^{4} 3^k$.
$3 + 9 + 27 + 81 = 120$.

28 Use $\sum^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{10} k$.
$\frac{10(11)}{2} = 55$.

Recursive sequences

29 A sequence satisfies $a_1 = 2$ and $a_{n+1} = a_n + 5$. Find a_5.
$a_1 = 2, a_2 = 7, a_3 = 12, a_4 = 17, a_5 = 22$.

30 A sequence satisfies $a_1 = 3$ and $a_{n+1} = 2a_n - 1$. Find a_4.
$a_1 = 3, a_2 = 5, a_3 = 9, a_4 = 17$.

31 A sequence satisfies $a_1 = 1$, $a_2 = 1$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_6.
$a_3 = 2, a_4 = 3, a_5 = 5, a_6 = 8$.

Identifying arithmetic and geometric sequences

32 State whether the sequence $4, 9, 14, 19, \ldots$ is arithmetic or geometric, and give the common difference or ratio.
Arithmetic, with common difference $d = 5$.

33 State whether the sequence $2, 6, 18, 54, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

Geometric, with common ratio $r = 3$.

34 State whether the sequence $100, 90, 80, 70, \ldots$ is arithmetic or geometric, and find its 10th term.

Arithmetic, $d = -10$. $a_{10} = 100 + 9(-10) = 100 - 90 = 10$.

Reading a sequence from its terms

35 The bar chart shows the first five terms of a sequence.

a Determine whether the sequence is arithmetic or geometric, and find the common ratio or difference. Geometric — each term is double the previous one, so $r = 2$.

b Find the 8th term. $a_8 = 2 \times 2^7 = 2 \times 128 = 256$.

Word problems: sequences in context

36 This question has two parts. An employee's starting monthly salary is 4000 dollars, and it increases by a fixed amount of 150 dollars each month.

a Write a formula for the salary a_n in month n (with $a_1 = 4000$). $a_n = 4000 + (n-1)(150)$.

b Find the salary in month 12. $a_{12} = 4000 + 11(150) = 4000 + 1650 = 5650$ dollars.

37 This question has two parts. A bacteria culture starts with 500 cells and triples every hour. Let a_n be the number of cells after n hours, with $a_0 = 500$.

a Write a formula for a_n. $a_n = 500 \times 3^n$.

b Find the number of cells after 4 hours. $a_4 = 500 \times 3^4 = 500 \times 81 = 40500$ cells.

MCQ — Arithmetic sequences: finding terms

38 Find the 14th term of the arithmetic sequence with $a = 6$, $d = 4$.

a. 62

b. 54

c. 50

d. 58 ✓

$T_{14} = 6 + 13(4) = 58$.

39 Find the 18th term of the arithmetic sequence with $a = -7$, $d = 3$.

a. 50

b. 41

c. 47

d. 44 ✓

$T_{18} = -7 + 17(3) = 44$.

40 Find the 12th term of the arithmetic sequence with $a = 80$, $d = -5$.

a. 20

b. 30

c. 25 ✓

d. -25

$T_{12} = 80 + 11(-5) = 25$.

41 Write the first four terms of the arithmetic sequence with $a = 9$, $d = -3$.

a. $9, 12, 15, 18$

b. $9, 6, 3, 0$ ✓

c. $9, 3, 0, -3$

d. $9, -3, -15, -27$

Each term subtracts 3: $9, 6, 3, 0$.

42 Find the 25th term of the arithmetic sequence with $a = 2$, $d = 7$.

a. 177

b. 170 ✓

c. 175

d. 163

$T_{25} = 2 + 24(7) = 170$.

MCQ — Arithmetic sequences: finding a and d

43 The 5th term of an arithmetic sequence is 19 and the 10th term is 44. Find the first term a and common difference d.

a. $a = -5, d = 5$

b. $a = -1, d = 4$

c. $a = 4, d = 5$

d. $a = -1, d = 5$ ✓

$d = \dfrac{44 - 19}{5} = 5$; $a = 19 - 4(5) = -1$.

44 An arithmetic sequence has $a_1 = 8$ and $a_7 = 32$. Find the common difference d.

a. $d = 4$ ✓

b. $d = 24$

c. $d = 4.8$

d. $d = 3$

$d = \dfrac{32 - 8}{6} = 4$.

45 An arithmetic sequence has $a_2 = 9$ and $a_6 = 29$. Find the common difference d.

a. $d = 5.5$

b. $d = 4$

c. $d = 20$

d. $d = 5$ ✓

$d = \frac{29 - 9}{4} = 5$.

46 The 3rd term of an arithmetic sequence is 15 and the 8th term is 45. Find the first term a and common difference d.

a. $a = 3, d = 5$

b. $a = 3, d = 6$ ✓

c. $a = 6, d = 3$

d. $a = 9, d = 6$

$d = \frac{45 - 15}{5} = 6$; $a = 15 - 2(6) = 3$.

MCQ — Arithmetic series: the sum formula

47 Find the sum of the first 25 terms of the arithmetic sequence with $a = 4$, $d = 3$, using $S_n = \frac{n}{2}(2a + (n - 1)d)$.

a. $1{,}075$

b. 960

c. $1{,}000$ ✓

d. 925

$S_{25} = \frac{25}{2}(8 + 24(3)) = 12.5(80) = 1{,}000$.

48 Find the sum of the first 18 terms of the arithmetic sequence with $a = 60$, $d = -4$.

a. 540

b. 456

c. 468 ✓

d. 480

$S_{18} = \frac{18}{2}(120 + 17(-4)) = 9(52) = 468.$

49 Find the sum of the arithmetic series $3 + 7 + 11 + \cdots + 59$.

a. 450

b. 434

c. 465 ✓

d. 480

$n = 15$ terms; $S_{15} = \frac{15}{2}(6 + 14(4)) = 7.5(62) = 465$.

50 Find the sum of the first 12 positive multiples of 7.

a. 504

b. 546 ✓

c. 462

d. 588

$7(1 + 2 + \cdots + 12) = 7(78) = 546.$

51 Find the sum of the first 30 terms of the arithmetic sequence with $a = 1$, $d = 2$ (the odd numbers).

a. 960

b. 930

c. 870

d. 900 ✓

$S_{30} = \frac{30}{2}(2 + 29(2)) = 15(60) = 900.$

MCQ — Geometric sequences: finding terms

52 Find the 7th term of the geometric sequence with $a = 2$, $r = 3$.

- **a.** 1,458 ✓
- **b.** 729
- **c.** 4,374
- **d.** 486

$T_7 = 2(3^6) = 1{,}458$.

53 Find the 6th term of the geometric sequence with $a = 200$, $r = 0.5$.

- **a.** 25
- **b.** 3.125
- **c.** 12.5
- **d.** 6.25 ✓

$T_6 = 200(0.5)^5 = 6.25$.

54 Find the 5th term of the geometric sequence with $a = 3$, $r = -2$.

- **a.** 24
- **b.** 48 ✓
- **c.** -24
- **d.** -48

$T_5 = 3(-2)^4 = 3(16) = 48$.

55 Write the first four terms of the geometric sequence with $a = 4$, $r = 2$.

a. $4, 6, 8, 10$

b. $4, 16, 32, 64$

c. $4, 8, 12, 16$

d. $4, 8, 16, 32$ ✓

Each term doubles: $4, 8, 16, 32$.

MCQ — Geometric sequences: finding a and r

56 The 2nd term of a geometric sequence is 8 and the 5th term is 64. Find r and a.

a. $r = 2$, $a = 8$

b. $r = 4$, $a = 2$

c. $r = 2$, $a = 4$ ✓

d. $r = 8$, $a = 1$

$r^3 = 64 \div 8 = 8 \Rightarrow r = 2$; $a = 8 \div r = 4$.

57 A geometric sequence has $a_1 = 5$ and $a_4 = 135$. Find r.

a. $r = 9$

b. $r = 27$

c. $r = 3$ ✓

d. $r = 2$

$r^3 = 135 \div 5 = 27 \Rightarrow r = 3$.

58 A geometric sequence has 3rd term 18 and 6th term 486. Find a and r.

a. $a = 18, r = 3$

b. $a = 6, r = 3$

c. $a = 2, r = 3$ ✓

d. $a = 2, r = 9$

$r^3 = 486 \div 18 = 27 \Rightarrow r = 3$; $a = 18 \div r^2 = 2$.

59 A geometric sequence has 2nd term 4 and 5th term -108. Find r.

a. $r = 3$

b. $r = -27$

c. $r = -3$ ✓

d. $r = -9$

$r^3 = -108 \div 4 = -27 \Rightarrow r = -3$.

60 A geometric sequence has $a_1 = 6$ and $a_4 = -162$. Find r.

a. $r = 3$

b. $r = -9$

c. $r = -3$ ✓

d. $r = -27$

$r^3 = -162 \div 6 = -27 \Rightarrow r = -3$.

MCQ — Geometric series: the finite sum

61 Find the sum of the first 7 terms of the geometric series with $a = 2$, $r = 2$, using $S_n = \dfrac{a(r^n - 1)}{r - 1}$.

a. 256

b. 508

c. 254 ✓

d. 127

$$S_7 = \frac{2(2^7 - 1)}{1} = 2(127) = 254.$$

62 Find the sum of the first 6 terms of the geometric series with $a = 200$, $r = 0.5$.

a. 196.875

b. 400

c. 396.875

d. 393.75 ✓

$$S_6 = \frac{200(1 - 0.5^6)}{0.5} = 393.75.$$

63 Find the sum of the first 5 terms of the geometric series with $a = 3$, $r = -2$.

a. 93

b. 63

c. -33

d. 33 ✓

$$S_5 = \frac{3((-2)^5 - 1)}{-2 - 1} = \frac{3(-33)}{-3} = 33.$$

64 Find the sum of the first 8 terms of the geometric series with $a = 1$, $r = 2$.

a. 255 ✓

b. 128

c. 254

d. 256

$$S_8 = \frac{1(2^8 - 1)}{1} = 255.$$

MCQ — Infinite geometric series

65 Find the sum of the infinite geometric series with $a = 6$, $r = \frac{1}{3}$.

a. 18

b. 6.75

c. 8

d. 9 ✓

$S = \frac{6}{1 - \frac{1}{3}} = \frac{6}{2/3} = 9.$

66 Find the sum of the infinite geometric series with $a = 12$, $r = \frac{1}{4}$.

a. 14

b. 16 ✓

c. 48

d. 15

$S = \frac{12}{1 - \frac{1}{4}} = \frac{12}{3/4} = 16.$

67 Find the sum of the infinite geometric series $9 - 3 + 1 - \frac{1}{3} + \cdots$.

a. 4.5

b. 9

c. 6

d. 6.75 ✓

$a = 9, r = -\frac{1}{3}$: $S = \frac{9}{1 + \frac{1}{3}} = \frac{9}{4/3} = 6.75.$

68 Express the repeating decimal $0.777\ldots$ as a fraction using an infinite geometric series with $a = 0.7$, $r = 0.1$.

a. $\frac{7}{90}$

b. $\frac{77}{100}$

c. $\frac{7}{9}$ ✓

d. $\frac{7}{10}$

$S = \frac{0.7}{1-0.1} = \frac{0.7}{0.9} = \frac{7}{9}.$

69 Find the sum of the infinite geometric series with $a = 10$, $r = \frac{2}{5}$.

a. $\frac{50}{3}$ ✓

b. $\frac{25}{3}$

c. 14

d. 25

$S = \frac{10}{1-\frac{2}{5}} = \frac{10}{3/5} = \frac{50}{3}.$

MCQ — Sigma notation

70 Evaluate $\sum_{k=1}^{6}(3k-1)$.

a. 51

b. 54

c. 57 ✓

d. 63

$2+5+8+11+14+17 = 57.$

71 Evaluate $\sum_{k=1}^{5} 2^k$.

a. 60

b. 62 ✓

c. 63

d. 32

$2 + 4 + 8 + 16 + 32 = 62.$

72 Use $\sum_{k=1}^{n} k = \frac{n(n+1)}{2}$ to evaluate $\sum_{k=1}^{15} k$.

a. 120 ✓

b. 110

c. 105

d. 136

$\frac{15(16)}{2} = 120.$

73 Evaluate $\sum_{k=2}^{5} k^2$.

a. 55

b. 30

c. 50

d. 54 ✓

$4 + 9 + 16 + 25 = 54.$

MCQ — Recursive sequences

74 A sequence satisfies $a_1 = 3$ and $a_{n+1} = a_n + 7$. Find a_5.

a. 31 ✓

b. 38

c. 28

d. 24

$3, 10, 17, 24, 31$: $a_5 = 31$.

75 A sequence satisfies $a_1 = 4$ and $a_{n+1} = 3a_n - 2$. Find a_4.

a. 10

b. 244

c. 28

d. 82 ✓

$4, 10, 28, 82$: $a_4 = 82$.

76 A sequence satisfies $a_1 = 1$, $a_2 = 2$, and $a_n = a_{n-1} + a_{n-2}$ for $n \geq 3$. Find a_7.

a. 21 ✓

b. 34

c. 8

d. 13

$1, 2, 3, 5, 8, 13, 21$: $a_7 = 21$.

77 A sequence satisfies $a_1 = 2$ and $a_{n+1} = 2a_n + 1$. Find a_4.

a. 11

b. 23 ✓

c. 5

d. 47

$2, 5, 11, 23$: $a_4 = 23$.

MCQ — Identifying arithmetic and geometric sequences

78 State whether $6, 11, 16, 21, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. **Arithmetic,** $d = 5$ ✓

b. Geometric, $r = \frac{11}{6}$

c. Geometric, $r = 5$

d. Arithmetic, $d = 6$

Each term adds 5: arithmetic, $d = 5$.

79 State whether $3, 9, 27, 81, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 6$

b. Arithmetic, $d = 6$

c. Arithmetic, $d = 3$

d. **Geometric,** $r = 3$ ✓

Each term is tripled: geometric, $r = 3$.

80 State whether $90, 80, 70, 60, \ldots$ is arithmetic or geometric, and find its 12th term.

a. **Arithmetic,** -20 ✓

b. Arithmetic, -10

c. Arithmetic, 30

d. Geometric, -20

$d = -10$: $T_{12} = 90 + 11(-10) = -20$.

81 State whether $5, 10, 20, 40, \ldots$ is arithmetic or geometric, and give the common difference or ratio.

a. Geometric, $r = 2$ ✓

b. Geometric, $r = 5$

c. Arithmetic, $d = 2$

d. Arithmetic, $d = 5$

Each term is doubled: geometric, $r = 2$.

MCQ — Reading a sequence from its terms

82 The bar chart shows the first five terms of a sequence. Find the common difference.

a. 6

b. 5 ✓

c. 4

d. 9

Each term increases by 5.

83 Using the same bar chart, predict the 8th term.

a. 29

b. 34

c. 39 ✓

d. 44

$d = 5$, $a = 4$: $T_8 = 4 + 7(5) = 39$.

84 Using the same bar chart, find the sum of the first five terms shown.

a. 70 ✓

b. 66

c. 24

d. 75

$4 + 9 + 14 + 19 + 24 = 70$.

MCQ — Word problems: sequences in context

85 An employee's starting monthly salary is 3,500 dollars, and it increases by 150 dollars each year. Find the salary in year 6.

a. 3,950 dollars

b. 4,400 dollars

c. 4,100 dollars

d. 4,250 **dollars** ✓

$a = 3{,}500$, $d = 150$: $T_6 = 3{,}500 + 5(150) = 4{,}250$.

86 A bacteria culture starts with 300 cells and doubles every hour. Find the number of cells after 5 hours.

a. 19,200

b. 1,500

c. 9,600 ✓

d. 4,800

$300 \times 2^5 = 300 \times 32 = 9{,}600$.

87 A theater's ticket sales grow according to a geometric sequence: 200 tickets sold on day 1, with a common ratio of 1.5. Find the number of tickets sold on day 4.

a. 900

b. 675 ✓

c. 300

d. 450

$T_4 = 200(1.5)^3 = 200(3.375) = 675.$